

ECON 311

Lecture 5: Modeling Economic Growth

A Production Model

The Solow Growth Model

Introduction

- To understand the causes of the great differences in production across time and countries, we first write down the following production function:

$$Y = F(K, L) = \bar{A}K^{\alpha}L^{(1-\alpha)} \quad , \quad \alpha \leq 1$$

- This production function has several theoretical properties that facilitates the analysis of empirical implications of differences in K and L .

Properties of the Production Function

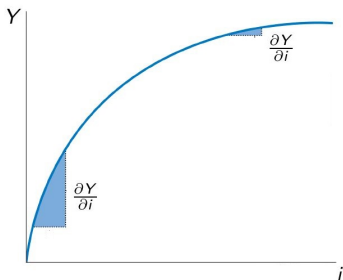
The Cobb-Douglas Production Function has the following properties:

1. Diminishing returns to K and L :

Diminishing Returns to i for $i = K, L$

$$MPK = \frac{\partial Y}{\partial K} = \alpha \bar{A} \left(\frac{L}{K} \right)^{(1-\alpha)}$$

$$MPL = \frac{\partial Y}{\partial L} = (1 - \alpha) \bar{A} \left(\frac{K}{L} \right)^{\alpha}$$



- Differences in K accumulation per person has different implications at different levels of development

Properties of the Production Function

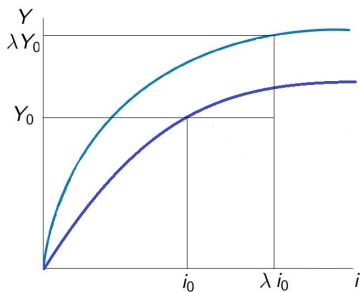
The Cobb-Douglas Production Function has the following properties:

2. Constant Returns to Scale:

$$F(\lambda K, \lambda L) = \lambda F(K, L)$$

$$F(\lambda K, \lambda L) = \lambda \underbrace{\bar{A} K^{\alpha} L^{(1-\alpha)}}_{F(K,L)}$$

Constant Returns to scale



- Population size matters for total income but does not for income per capita

Solving for Production Parameters

- To take this model to the data, we need ...
 - ▶ to assign value to the parameter, α
 - ▶ and study how differences in the exogenous variables, K , L , $\bar{A}$, result in differences in endogenous variable, Y .
- It turns out that from the optimal decision of firms to rent K and hire L , we obtain:

$$\frac{wL^*}{Y^*} = (1 - \alpha) \quad , \quad \frac{rK^*}{Y^*} = \alpha$$

- We can obtain this from the income approach to measuring GDP:
 $\alpha = \frac{1}{3}$

Development Accounting

- Our main concern is to measure GDP per capita differences, so we expressed the production in per capita terms:

$$y = \bar{A}k^{\frac{1}{3}}, \quad y = \frac{Y}{L}, \quad k = \frac{K}{L}$$

- We are going to assume that a country uses all of its resources to produce:

$$k = \bar{k}, \quad \bar{k} = \frac{\bar{K}}{\bar{L}}$$

- We observe differences in $\bar{k}$ but, because we do not observe differences in $\bar{A}$,...

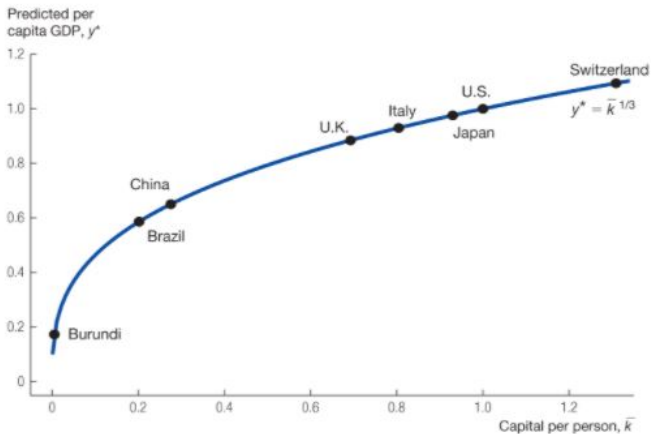
► we will assume that it is the same for all countries.

- In the following exercise, we will calculate the output of country c relative to the US :

$$\frac{y_c}{y_{US}} = \left(\frac{\bar{k}_c}{\bar{k}_{US}} \right)^{\frac{1}{3}}$$

Development Accounting

Predicted Per Capita GDP in the Production Model



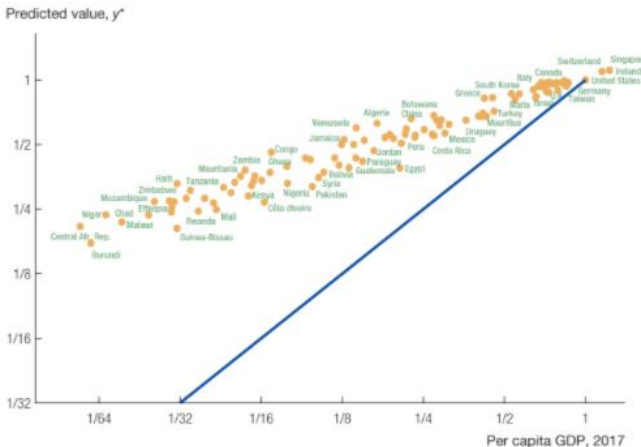
Development Accounting

The Model's Prediction for Per Capita GDP (U.S. = 1)

Country	Observed capital per person, $\bar{k}$	Predicted per capita GDP $y = \bar{k}^{1/3}$	Observed per capita GDP
U.S.	1.000	1.000	1.000
Switzerland	1.309	1.094	1.151
Japan	0.930	0.976	0.734
U.K.	0.693	0.885	0.714
Italy	0.804	0.930	0.680
Spain	0.731	0.901	0.640
China	0.276	0.651	0.279
Brazil	0.202	0.587	0.252
South Africa	0.174	0.559	0.214
India	0.081	0.433	0.117
Burundi	0.005	0.173	0.015

Development Accounting

The Model's Prediction for Per Capita GDP (U.S. = 1)



Model with only k and constant $\bar{A}$

- The model without $\bar{A}$ differences underestimate differences in output per capita.
- On average, y in industrialized countries is 10 times larger than in non-industrialized countries:

$$10 = \frac{\bar{A}^{ind} (k^{ind})^{\frac{1}{3}}}{\bar{A}^{nind} (k^{nind})^{\frac{1}{3}}}$$

- ▶ If we assume that $\bar{A}^{ind} = \bar{A}^{nind}$, then ...

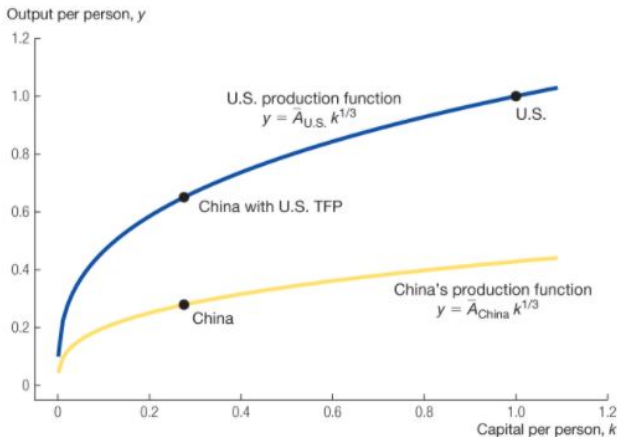
$$10^3 = 1000 = \frac{k^{ind}}{k^{nind}}$$

- ▶ There is no empirical evidence of such large differences.
- Presumably, differences in $\bar{A}$ play a big role in accounting for differences in y .
 - ▶ We cannot be certain of this because differences in $\bar{A}$ are unobserved in general.

Understanding differences in $\bar{A}$

- We could assume that our model is correct and interpret deviations of predicted from observed as differences in $\bar{A}$.

The U.S. and Chinese Production Functions



Understanding differences in $\bar{A}$

Measuring TFP So the Model Fits Exactly

Country	Per capita GDP (y)	$\bar{k}^{1/3}$	Implied TFP ($\bar{A}$)
United States	1.000	1.000	1.000
Switzerland	1.151	1.094	1.052
U.K.	0.714	0.885	0.807
Japan	0.734	0.976	0.752
Italy	0.680	0.930	0.731
Spain	0.640	0.901	0.710
China	0.279	0.651	0.429
Brazil	0.252	0.587	0.429
South Africa	0.214	0.559	0.383
India	0.117	0.433	0.270
Burundi	0.015	0.173	0.084

Calculations are based on the equation $y = \bar{A}\bar{k}^{1/3}$. Implied productivity $\bar{A}$ is calculated from data on y and $\bar{k}$ for the year 2010, so that this equation holds exactly as $\bar{A} = y/\bar{k}^{1/3}$.

Understanding differences in $\bar{A}$

- A common interpretation of differences in $\bar{A}$ is differences in Total Factor Productivity (TFP)
 - ▶ Some countries have high y because of large reserves of natural resources (e.g. oil rich countries), but usually is due to large factor productivity.
- Important factors that have been shown to account for TFP differences across countries are:
 1. Differences in Human Capital
 2. Differences in Technology
 3. Differences in Institutions that promote economic growth
 4. Missallocation of production resources

A Production Model

The Solow Growth Model

Introduction

- The production model is very useful because it is able to display two features of the data:

1. $\frac{y_{ind}}{y_{nind}} = \frac{\bar{A}_{ind}(\bar{k}_{ind})^{\frac{1}{3}}}{\bar{A}_{nind}(\bar{k}_{nind})^{\frac{1}{3}}} \text{ (differences in levels of } y\text{)}$

2. $g_{y_{ind}} - g_{y_{nind}} = (g_{\bar{A}_{ind}} + \frac{1}{3}g_{\bar{k}_{ind}}) - (g_{\bar{A}_{nind}} + \frac{1}{3}g_{\bar{k}_{nind}}) \text{ (differences in growth rates of } y\text{)}$

- Nevertheless, it does not provide any explanation as to why the growth rates are so different across countries.
 - ▶ The Solow Model provides an explanation for K accumulation, which will help us to explain why capital per worker grows at a faster pace in some countries.
 - ▶ As we have seen this is only a part of the explanation, but it is a very useful starting point for understanding the big differences in growth rates we observe in the data.

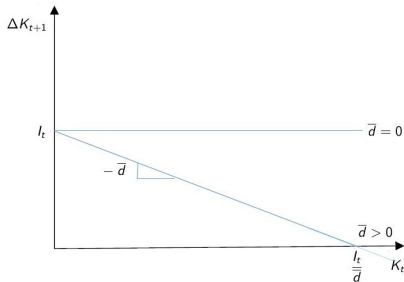
Capital Accumulation

- At the center of the Solow Model is the capital accumulation equation:

$$K_{t+1} = I_t + (1 - \bar{d})K_t$$

- I =investment, $\bar{d}$ =depreciation rate ($0 \leq \bar{d} \leq 1$)
- We can rewrite this equation in changes: $\Delta K_{t+1} = K_{t+1} - K_t$:

$$\Delta K_{t+1} = I_t - \bar{d}K_t$$



Capital Accumulation

- Investing in capital for $t + 1$ requires resources at t :

$$C_t + I_t = Y_t \quad (\text{resource constraint})$$

- ▶ An assumption of the Solow model is that capital and consumption goods are produced using the same production function.
- We are going to assume that individual decisions to save and invest is exogenous:

$$I_t = \bar{s}Y_t \quad (\text{allocation of resources}) \quad , \quad 0 \leq \bar{s} \leq 1$$

- Combining the allocation of resources with the capital accumulation equation, we obtain:

$$\Delta K_{t+1} = \bar{s}Y_t - \bar{d}K_t \quad (\text{law of motion})$$

Capital Accumulation

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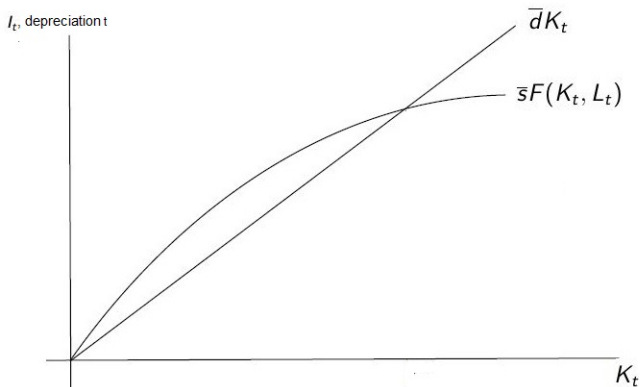
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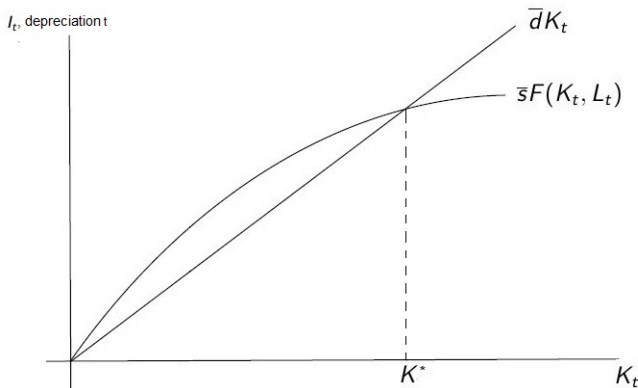
$$\Delta K_{t+1} = \bar{s}F(K_t, L_t) - \bar{d}K_t \quad (\text{law of motion})$$

Dynamics of Capital Accumulation



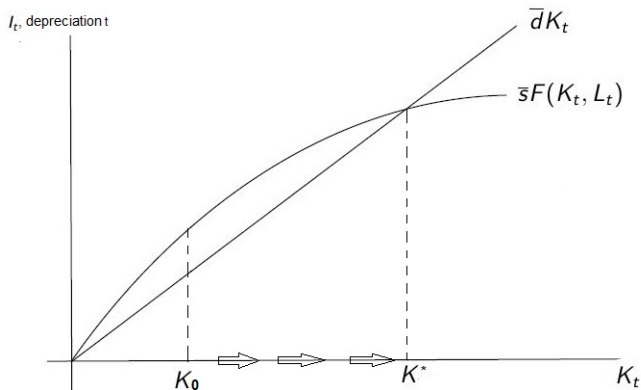
$$\Delta K_{t+1} = sF(K_t, L_t) - \bar{d}K_t$$

Dynamics of Capital Accumulation



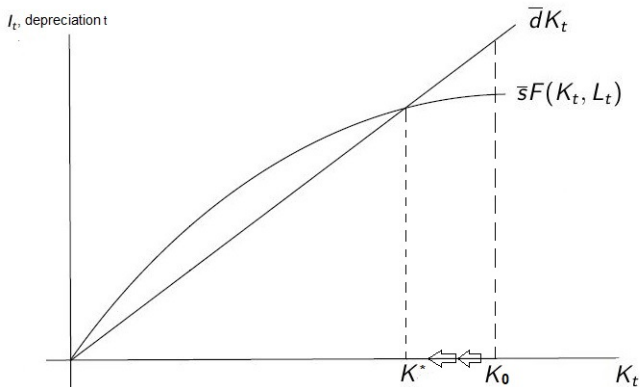
$$\underbrace{\Delta K_{t+1}}_{K_{t+1}=K_t} = 0 \Leftrightarrow \underbrace{\bar{s}F(K_t, L_t)}_{I_t} = \bar{d}K_t$$

Dynamics of Capital Accumulation



$$\underbrace{\Delta K_{t+1}}_{K_{t+1} > K_t} > 0 \Leftrightarrow \underbrace{\bar{s}F(K_t, L_t)}_{I_t} > \bar{d}K_t$$

Dynamics of Capital Accumulation



$$\underbrace{\Delta K_{t+1}}_{K_{t+1} < K_t} < 0 \Leftrightarrow \underbrace{\bar{s}F(K_t, L_t)}_{I_t} < \bar{d}K_t$$